Exploratory Data Analysis & Insights on Student Performance

MINOR PROJECT REPORT

Submitted by: Ayush Roy

Domain: Data Science

Institution: Skillairo Edutech Pvt Ltd

1. Introduction

Educational institutions collect a large amount of student data, but often it is not used to get useful insights. The purpose of this project is to perform Exploratory Data Analysis (EDA) on the Student Performance dataset to find patterns that affect student performance. EDA helps us understand the data, check its quality and see how variables relate to each other. Using Python libraries like Pandas, NumPy, Matplotlib and Seaborn, the data was cleaned and analysed with univariate, bivariate and multivariate analysis. Graphs were used to see how factors like study time, previous failures, absences and earlier grades are linked to the final grade, and the findings are summarised in plain language.

1.1 Problem Statement

Educational institutions collect large volumes of student data but fail to extract actionable insights.

1.2 Objectives

- Understand the dataset structure and variables.
- Check and prepare the data for analysis.
- Use univariate, bivariate and multivariate analysis to investigate patterns.
- Visualise findings and summarise useful insights in plain language.

2. Dataset Overview

Dataset: Student Performance — UCI Machine Learning Repository

https://archive.ics.uci.edu/ml/datasets/Student+Performance

In this EDA, we have analysed the student-mat.csv file, containing 395 rows and 33 columns.

3. Data Understanding and Preprocessing

3.1 Initial Inspection

The dataset was initially inspected using Python's Pandas library to understand its structure and shape. The dataset contains 395 rows and 33 columns, where each row represents a separate student.

The df.info() function was used to check column names and their data types and to check for non null values.

The dataset contains information about students' personal details like their family background, their families' support for them and their study habits. Some important variables include age, studytime, failures and absences. The columns G1, G2, G3 represents the students' first period, second period and final grades respectively.

Initial inspection also revealed that columns had different data types including integers, strings or object type values. Some integer columns used integers to represent categories rather than continuous numerical values. The object-type values and string values were all converted into integer data type columns.

3.2 Data Quality Checks

Using df.info(), we find that there isn't any null rows in all of the 395 rows. So there is no removal of rows containing null columns. We haven't used outliers removal, as such outlier values could provide important insights during the analysis.

4. Univariate Analysis

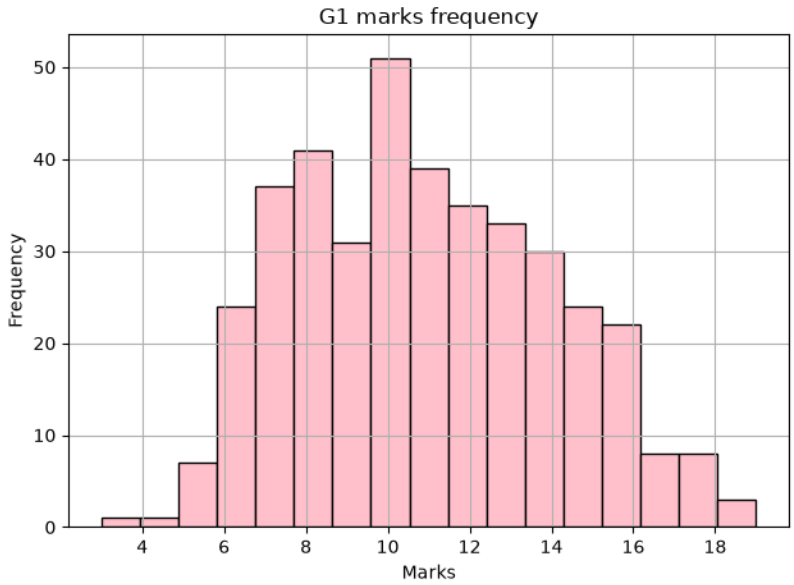
This section examines individual variables to understand their distributions, typical values and unusual observations.

4.1 First-Period Grade (G1)

From the histogram we can see that the typical grade range is from 3 to 19.

Most of the students have their marks spread between the 6-16 marks range and there is a slight increase in students at marks 10.

No students has got the 0 mark, while no students got full marks either.



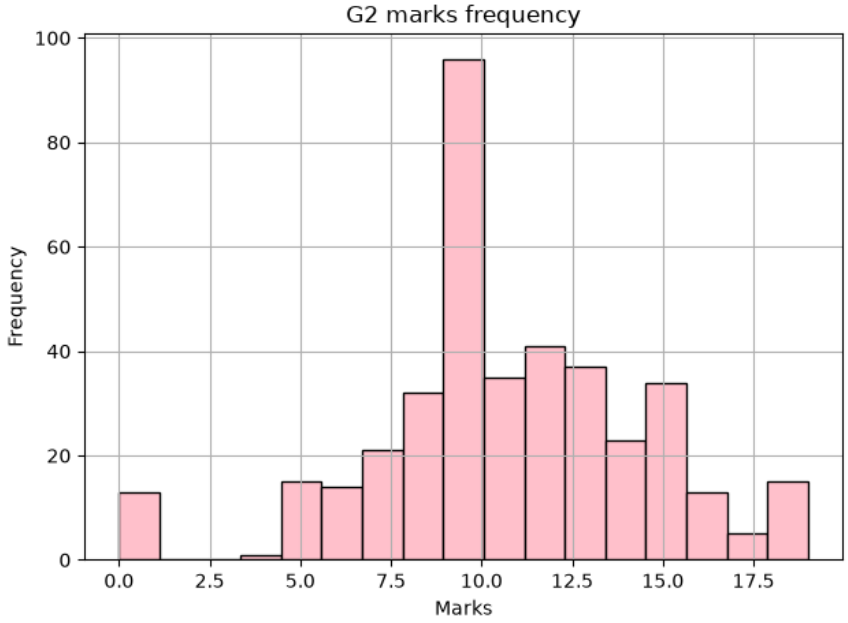
Histogram Graph of G1

4.2 Second-Period Grade (G2)

From the histogram we can see that the typical grade range is from 0 to 19.

There is a sharp increase of students in the 9-10 marks range a nearly similar distribution from 8.5 to 16.5 marks range.

There are students who have got the 0 marks but no students have secured full marks.



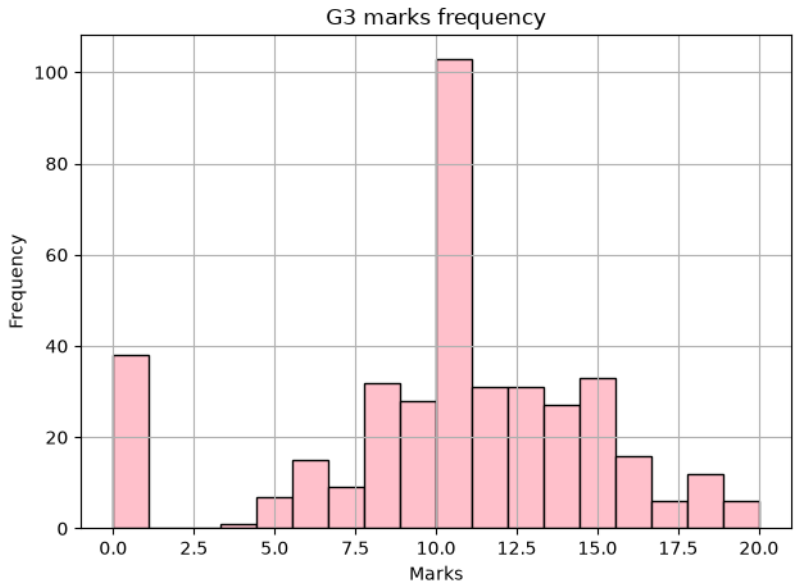
Histogram Graph of G2

4.3 Final Grade (G3)

From the histogram we can see that the typical grade range is from 0 to 20.

Most students have their marks in the 8.5 -15 marks range with a noticeable dip at the lower end and with a sharp increase of students who secured 9-11 marks.

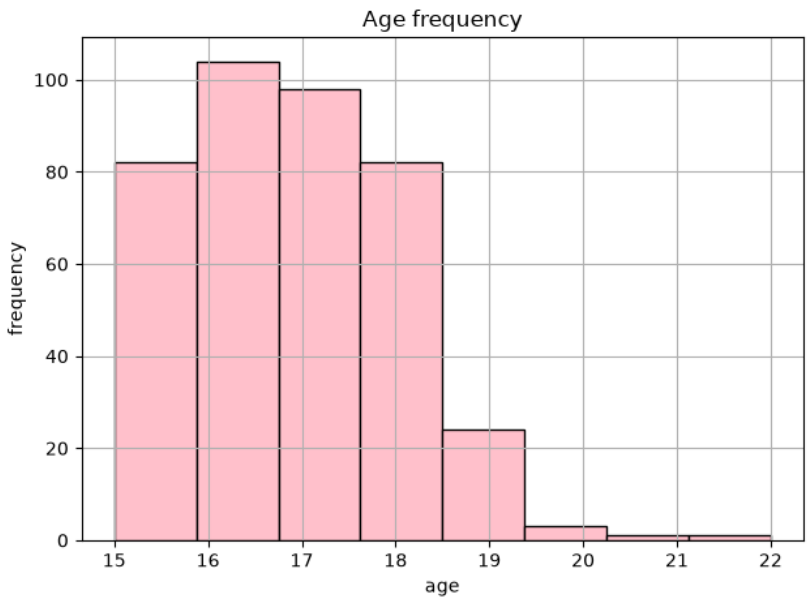
There is also a sudden increase in students who scored 0 marks and very few students have achieved full marks as well.



Histogram Graph of G3

4.4 Age

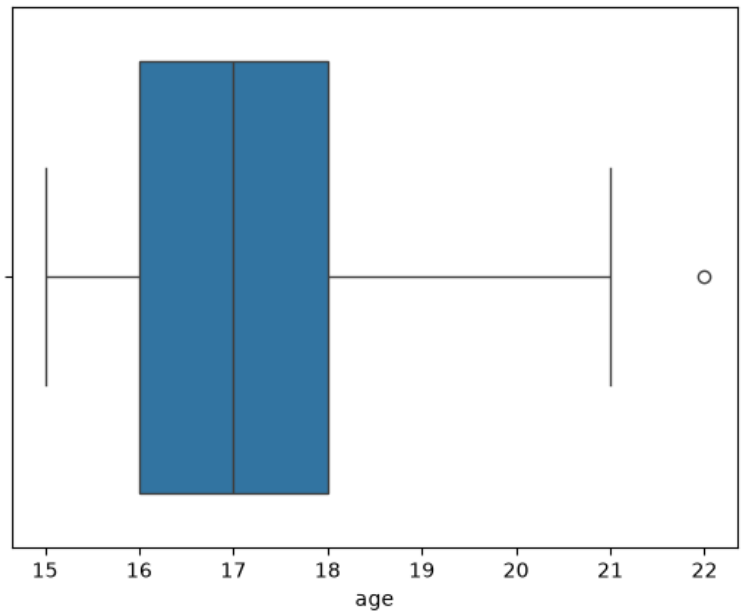
From the histogram we see that 15-18 age range is the most common and very few students have age above 19.



Histogram Graph of Age

From the boxplot we see 17 is the median age, 25% students are 16 years or younger and 75% students are 18 years or younger .

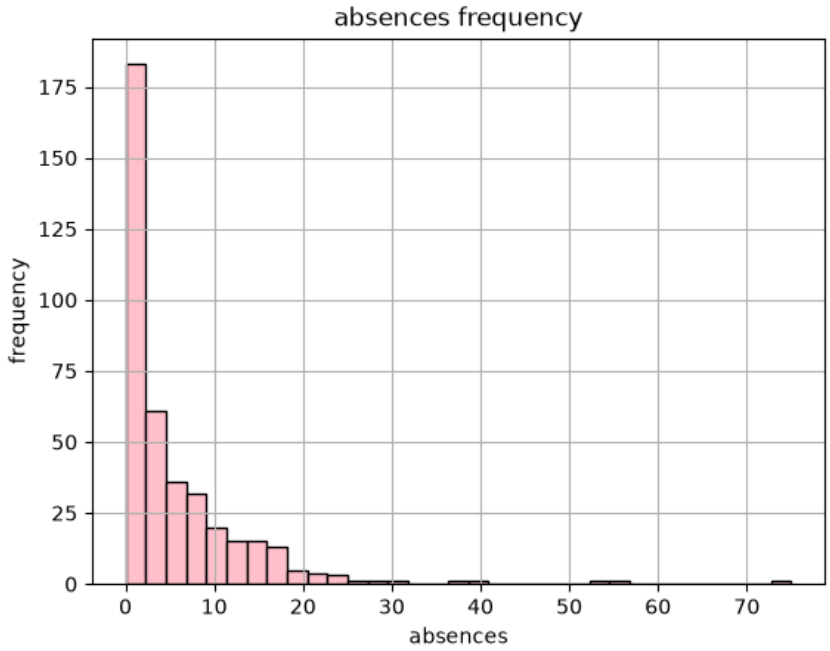
15 is the lowest non outlier and 21 is the highest non outlier.



Boxplot Graph of Age

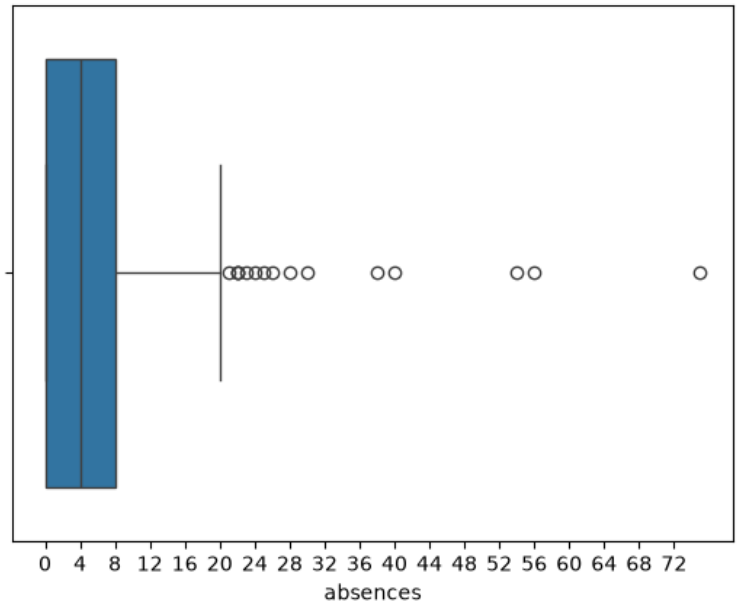
4.5 Absences

From the histogram, we see most students have 0 absences and the frequency of students with absences higher than that progressively reduces.



Histogram Graph of Absences

From the boxplot, we see the median of absences is 4 and 75% of students have 8 absences or less and 20 is the largest non outlier.



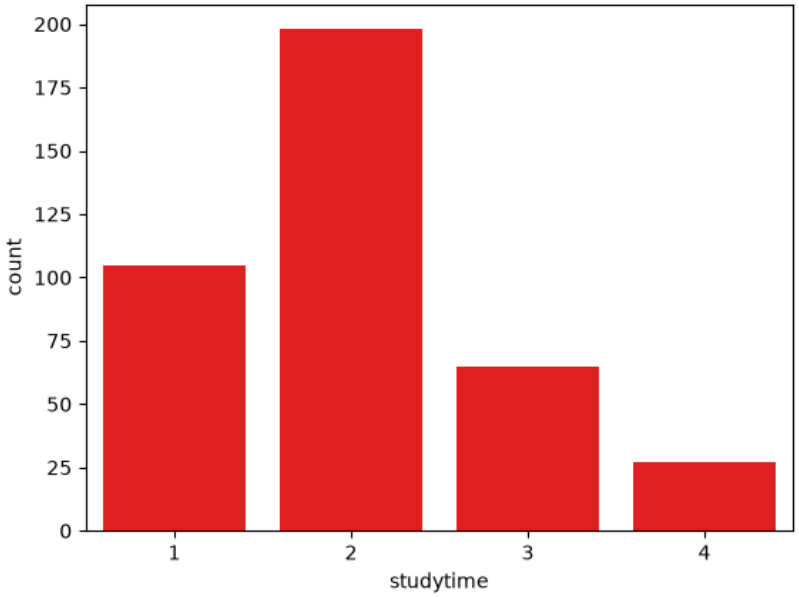
Boxplot Graph of Absences

4.6 Study Time

From the countplot, we see that category 2 is the most common, about 50% of all students.

Category 4 is the least common.

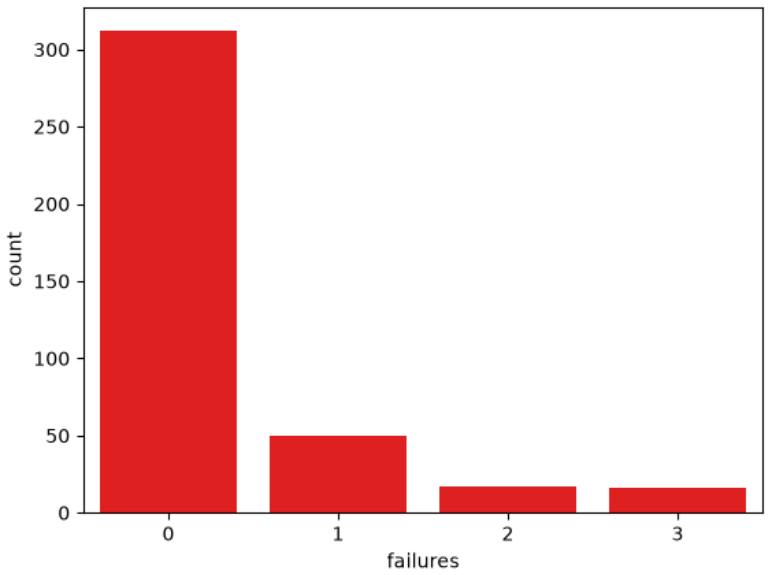
Around 23.2% of all students study more than 5 hours a week.



Bar Graph of Studytime

4.7 Previous Failures

We see from the countplot around 79% of students have 0 past failures and remaining has 1 or more previous failures. Category 3 is the least common.



Bar Graph of Failures

4.8 Univariate Summary

• Most students have scored between 6-16 marks in G1, while G2 and G3 have showed a noticeable concentration around 9-11 marks.

• Most students belong to the 15-18 age group. The median age is 17 years, with 75% of students being 18 years or younger.

• The frequency distribution shows that students with zero previous failures form the largest group, while students with multiple previous failures are considerably fewer.

• Studytime category 2 is the most common, accounting for approximately 50% of students, whereas category 4 is seen to be the least common.

5. Bivariate Analysis

This section examines how selected student characteristics and earlier-period grades are associated with students' final grades (G3), using boxplots and scatterplots.

5.1 Study Time vs Final Grade (studytime vs G3)

Question: Do final-grade distributions differ across study-time categories?

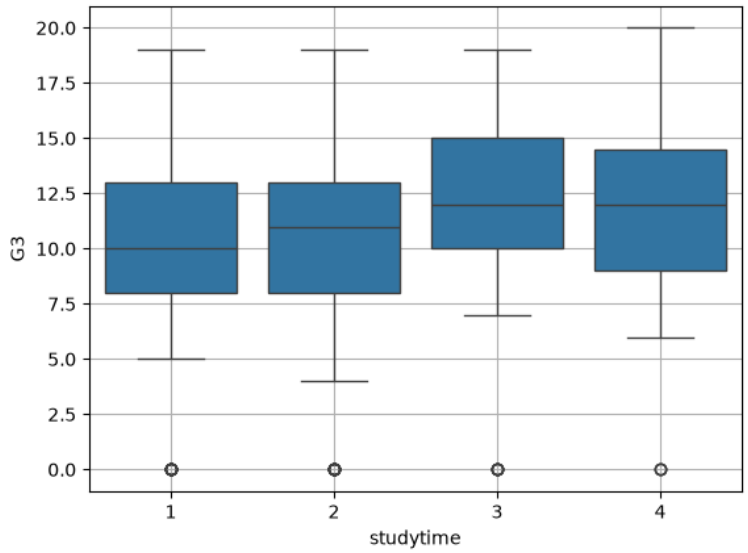
Observed pattern:

• From the boxplot we see that median marks in G3 generally increase with increase in studytime category 1 to 3.

• Category 3 and categorie 4 have approx same median final marks, about 12 marks.

• The distribution across categories overlap considerably, showing students with diff studytime can have similar marks.

• Each studytime category has a lower outlier at 0.
• Greater studytime tends to show a higher median marks even though this trend is not consistent and the overlap of distribution across the categories show students with different study hours can have similar marks.

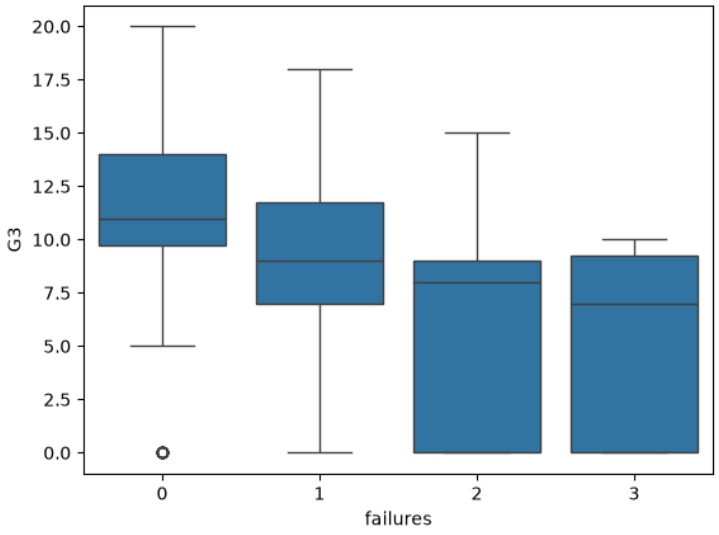


Boxplot Graph Studytime vs G3

5.2 Previous Failures vs Final Grade (failures vs G3)

Question: Are previous failures associated with differences in final grades?

Observed pattern:
• From the boxplot we see that the median final marks keeps going down with increase in failures category 0 to 3 and the trend is consistent.
• Overlapping of distribution is significant in failures 2 and 3 which shows at higher failure categories, student performances start to become similar.
• The highest non outlier keeps on decreasing with increase in failures categories as well.



Boxplot Graph of Failures vs G3

5.3 Absences vs Final Grade (absences vs G3)

Question: Is the number of absences associated with final grades?

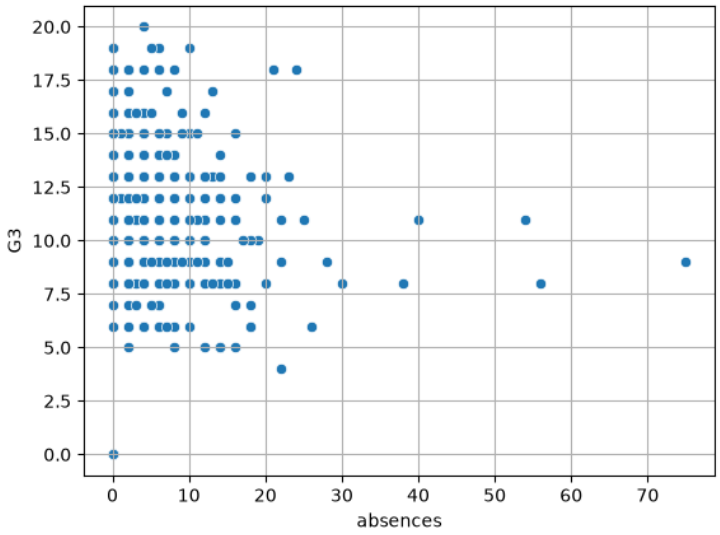
Observed pattern:

• Most students have less than 20 absences.

• Students with fewer absences show a wide range of marks attained.

• A few students have unusually high absences but having higher absences has no as such effects on the final marks as they do not achieve the lowest grades.

• There is no visible correlation in the scatterplot.



Scatterplot Graph of Absences vs G3

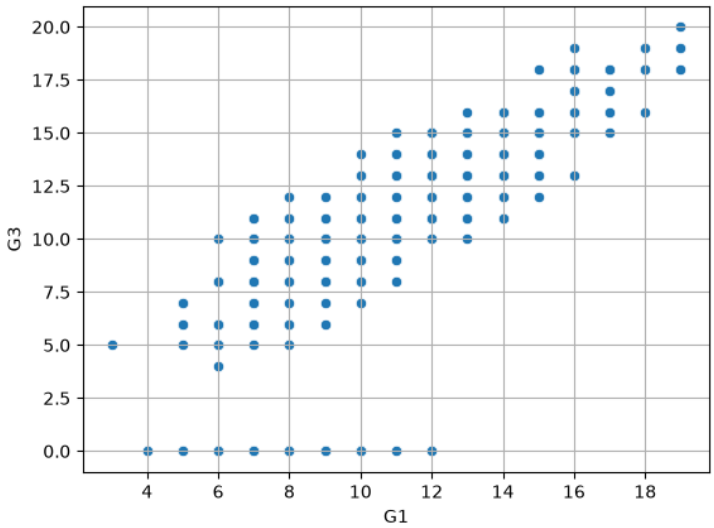
5.4 First-Period Grade vs Final Grade (G1 vs G3)

Question: How strongly are G1 and G3 associated?

Observed pattern:

• From scatterplot, we see a strong positive relationship between G1 and G3 as the G3 marks gradually increase with increase in G1 marks.

• Some students from G1 marks 4 to 12 have secured 0 marks in G3 and seem out of the general relationship we are seeing.



Scatterplot Graph of G1 vs G3

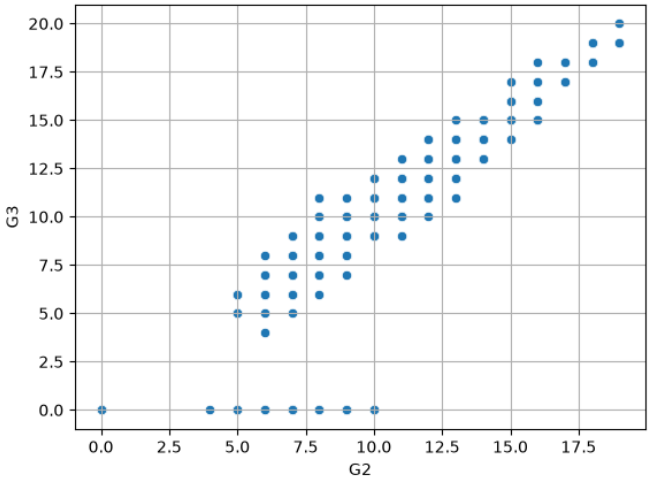
5.5 Second-Period Grade vs Final Grade (G2 vs G3)

Question: How strongly are G2 and G3 associated?

Observed pattern:

• From scatterplot we see a more stronger positive relationship b/w G2 & G3 than we saw in G1 & G3 as the spread is lesser.

• Just like previous one, there are some students who have scored between 3 and 10 marks in G2 but has got 0 marks in G3.



Scatterplot Graph of G2 vs G3

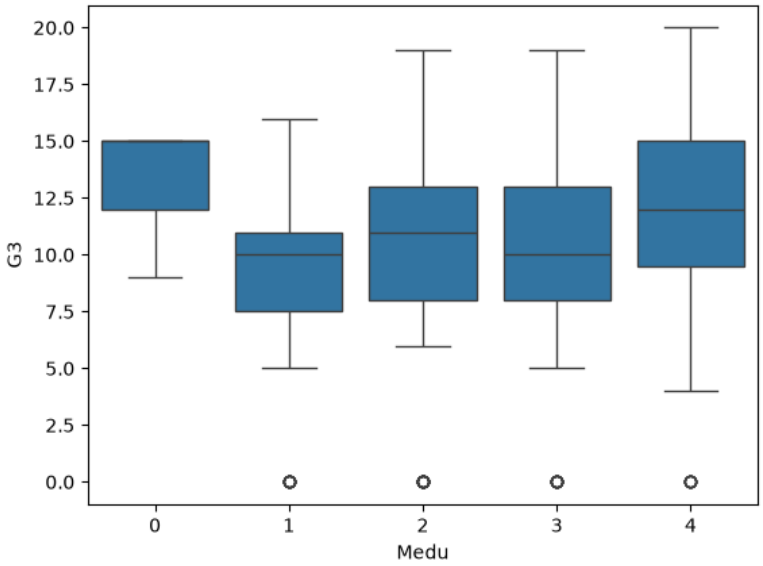
5.6 Mother's Education vs Final Grade (Medu vs G3)

Question: Do final-grade distributions differ by mother's education level?

Observed pattern:

• From the boxplot we see that median marks in G3 does not generally increase or decrease with increase in Mother's education, while the median marks for Medu category 0 is the highest and category 4 is the second highest.

• For Medu category 0, median appears to coincide with the third quartile, meaning both values are approximately equal. There are no upper outliers observed for this category.



Boxplot Graph of Medu vs G3

5.7 Weekday Alcohol Consumption vs Final Grade (Dalc vs G3)

Question: Is weekday alcohol consumption associated with final grades?

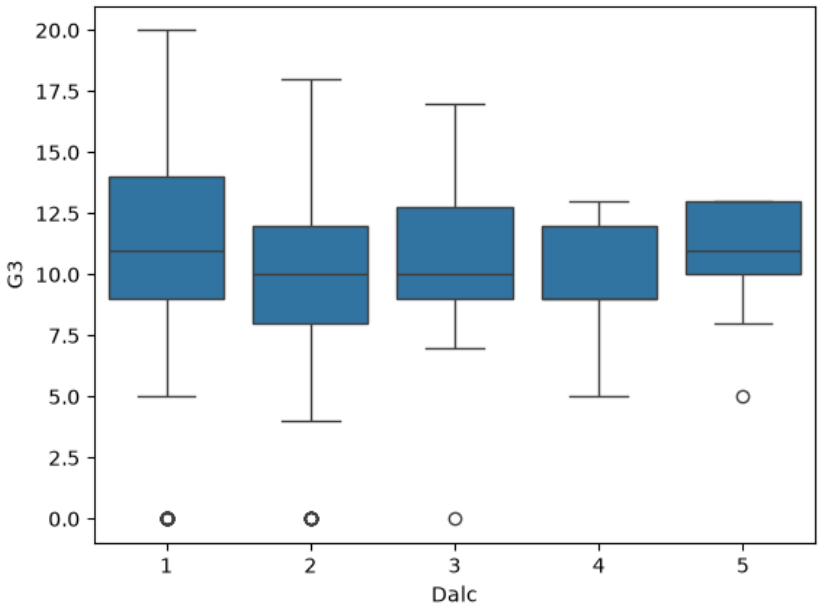
Observed pattern:

From the boxplot we see that median marks in G3 seem to have a slight decrease with increase in Dalc category though this trend is not followed by Dalc category 5.

• For Dalc category 4, the median seems to coincide with the first quartile (25%).

• And there is no upper non outlier bound for Dalc category 5.

• The distribution across categories overlap significantly, showing students with different weekday alcohol consumption can have similar marks.



Boxplot Graph of Dalc vs G3

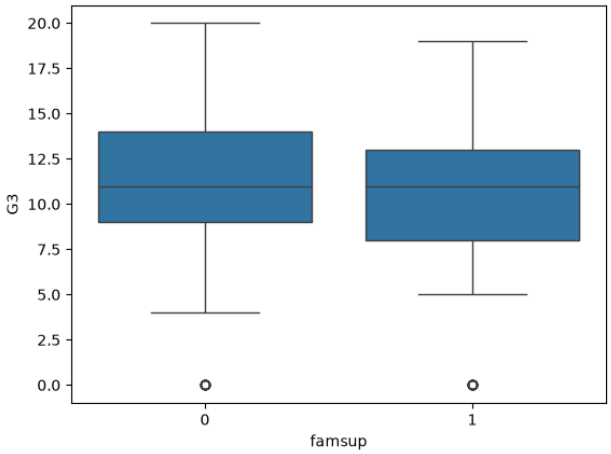
5.8 Family Educational Support vs Final Grade (famsup vs G3)

Question: Do final-grade distributions differ according to reported family educational support?

Observed pattern:

• We see the median marks approximately equal for both famsup category.

• While the marks distribution overlap significantly, students without family support is seen to have higher third quartile and first quartile than students with family support.



Boxplot Graph of Famsup vs G3

5.9 Bivariate Summary

• Study time: There appears to be a positive relationship between the amount of hours students spend studying and their median final grade. However, there is a lot of overlap meaning that students who study different amounts have similar final grades.

• Previous failures: There is a clear negative relationship between the amount of previous failures and the median final grade. The highest value for each category (not excluding outliers) is decreasing meaning there is a noticeable relationship between previous failures and final grade.

• Absences: Most students have less than 20 absences. However those with less absences have a large spread in their grades. From the scatterplot we can see there is no visible correlation between the amount of absences and final grade.

• G1 and G2: Both G1 and G2 appear to have a strong positive relationship with G3. However G2 appears to have a stronger relationship with G3 as the spread is smaller.

• Other factors (Medu, Dalc and Famsup): There is no clear increasing/decreasing relationship between the mothers education and the median final grade. Those who consume alcohol on weekdays have a slight negative relationship with their median grade. However this isn't clear. There is a lot of overlap in the distribution of grades for family educational support. The median grade is roughly the same for both categories.

6. Multivariate Analysis

This section examines combinations of three variables to explore how study time, previous failures and absences relate to final grades, and how the first-period and second-period grades correlate with the final grade.

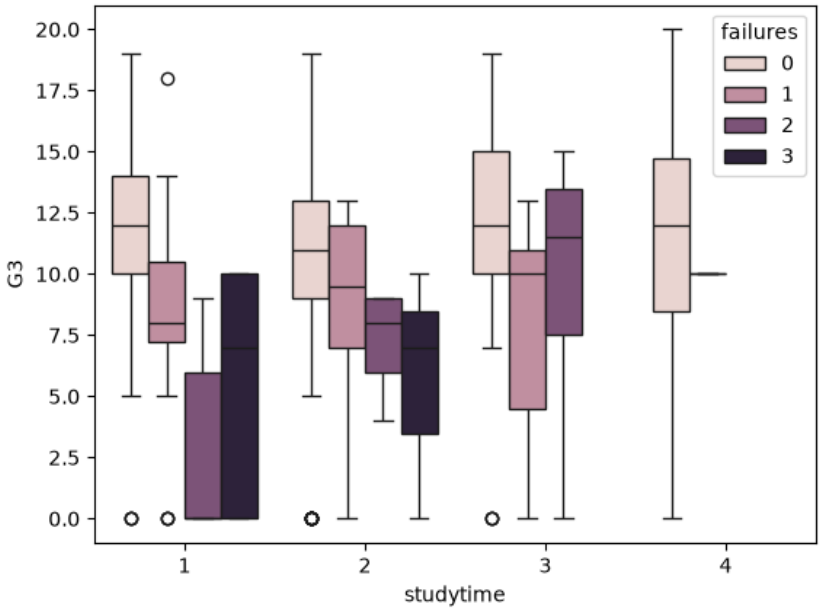
6.1 Study Time, Previous Failures and Final Grade

Question and method: How do final-grade distributions vary across study-time and previous-failure categories? A grouped boxplot was used to compare G3 across these categories.

Finding: • Students with 0 failures have higher median marks than the other failure groups.

• For study-time categories 1 and 2, median marks decrease as the failure category increases.

• Study-time categories 3 and 4 have very few students in some failure groups; therefore, these groups should be interpreted cautiously.



Boxplot Graph of G3 by Study Time and Previous Failures

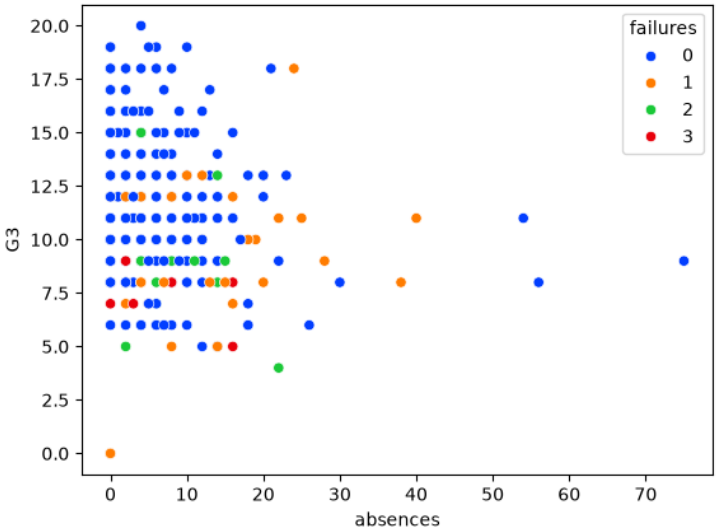
6.2 Absences, Failures and Final Grade

Question and method: How do absences and previous failures relate to final grades? A scatterplot was used to compare absences with G3, with points distinguished by previous-failure category.

Finding: • Most students have fewer than 20 absences, and most students have 0 previous failures.

• Students with 0 failures show a wide range of marks, including the highest marks.

• While there is no clear relationship, higher failures are generally associated with lower final grades.



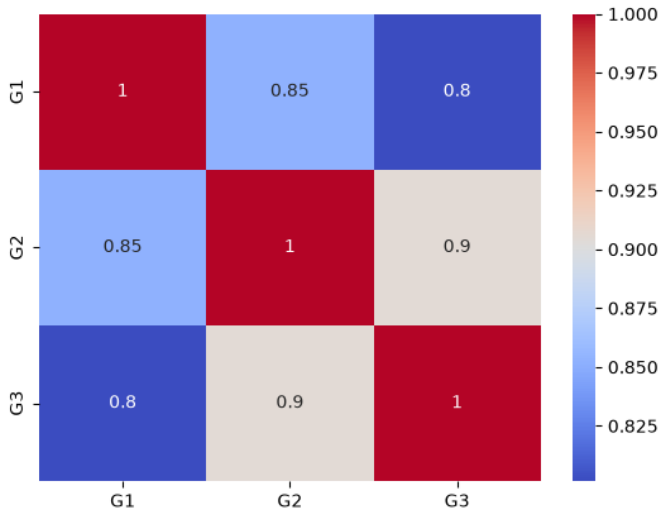
Scatterplot Graph of Absences vs G3 by Previous Failures

6.3 First-Period, Second-Period and Final Grade

Question and method: How strongly are G1, G2 and G3 correlated? A correlation heatmap was used to compare the correlation coefficients among the three grade variables.

Finding: • G1 has a correlation of 0.85 with G2 and 0.8 with G3, while G2 has a correlation of 0.9 with G3.

• Therefore, G1, G2 and G3 show strong correlations, with the strongest correlation being between G2 and G3.



Correlation Heatmap of G1, G2 and G3

7. Key Insights

1. G1 and G2 are strongly related to the final grade. G1 has a correlation of 0.8 with G3 and G2 has 0.9, and the G2 scatterplot has less spread. This shows that how a student does early on is closely linked to how they finish, so these grades can help spot students who may need extra help.

2. Previous failures show the clearest negative pattern with G3. The median final grade goes down from failure category 0 to 3, and the highest non-outlier mark also goes down. So students who have failed before may need attention earlier.

3. Study time has a positive but weak relationship with final grades. The median G3 goes up from category 1 to 3, but the boxes overlap a lot and categories 3 and 4 have about the same median of 12. Study time alone does not decide a student's marks.

4. Absences do not show a clear relationship with final grades. Most students have less than 20 absences, and those with few absences have marks all over the range. So absences alone are not a good sign of performance in this data.

5. Family and lifestyle factors show weak or unclear patterns. Mother's education has no steady trend with G3, weekday alcohol use shows only a slight and uneven drop in median marks, and both famsup groups have about the same median. On their own these seem less useful than a student's earlier results.

6. Some students scored 0 in G3 even though their G1 and G2 marks were moderate (about 3 to 12). They do not follow the strong positive relationship between the grades, so they are worth looking at separately. They may be dropouts or missing assessments.

8. Limitations

- The dataset represents a particular study context; findings may not generalise to all schools or students.
- Observed associations do not establish causation.
- The analysis uses only the student-mat.csv file, which has 395 students and covers just the Maths course, so the sample is fairly small.
- Only visual and descriptive methods were used (histograms, boxplots, scatterplots and correlations). No statistical tests or prediction models were used to confirm the patterns.
- Some categories have very few students, such as the higher failure and study-time groups, so patterns there should be treated carefully.
- Outliers like the students with 0 in G3 were kept, and they may affect some of the patterns.
- G1 and G2 are strongly correlated with G3 and with each other, so it is hard to separate what each one contributes.

9. Conclusion

In this project, EDA was used on the Student Performance dataset (395 students, 33 columns) to find patterns linked to final grades. There were no missing values, so univariate, bivariate and multivariate analysis was done using histograms, boxplots, scatterplots and a correlation heatmap. The results show that earlier grades are the strongest sign of the final grade, especially G2, which has a correlation of 0.9 with G3. Previous failures are clearly linked to lower final grades. Study time shows a small positive pattern, while absences, mother's education, alcohol use and family support show weak or unclear relationships. These are associations and not causes, but they show that EDA can point out which factors, like early grades and past failures, are worth looking at when supporting students.

References

Cortez, P. and Silva, A. (2008). Using data mining to predict secondary school student performance. Student Performance dataset, UCI Machine Learning Repository.

Dataset link: https://archive.ics.uci.edu/ml/datasets/Student+Performance